

A Brief Introduction to Point Distortion Model

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The point distortion model is a projective model for those points in the 3-D scene would be projected on a 2-D image plane through the distorted imaging system, instead of a pinhole, through which a non-distorted image would be acquired. Relation between the non-distorted image and the distorted image can be described by the following formula, such that,

$$\begin{pmatrix} x_d \\ y_d \end{pmatrix} = (1 + k_1(x^2 + y^2) + k_2(x^2 + y^2)^2) \begin{pmatrix} x \\ y \end{pmatrix}$$

where (x_d, y_d) is the distorted point coordinate, and (x, y) is the non-distorted one.

The program to calculate the distorted image can be divided into 3 parts. The first part is to calculate $(1 + k_1(x^2 + y^2) + k_2(x^2 + y^2)^2)$, which is given by

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1 k_matrix = 1+k(1)*(x_o.^2+y_o.^2)+k(2)*(x_o.^2+y_o.^2).^2;
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We can calculate the corresponding non-distorted coordinate (x_m, y_n) of integer coordinate (m, n) in distorted image by

$$\begin{pmatrix} x_m \\ y_n \end{pmatrix} = \frac{1}{(1 + k_1(x^2 + y^2) + k_2(x^2 + y^2)^2)} \begin{pmatrix} m \\ n \end{pmatrix}$$

It is noticed that (x_m, y_n) may have decimal parts which would not fit in one certain integer point. So we add a interpolation based on the area divided by the distorted coordinate into the integer points around, as is shown as follows, where $(i, j), (i+1, j), (i, j+1)$ and $(i+1, j+1)$ are integer points around (x_m, y_n) with $i = \lfloor x_m \rfloor$, and $j = \lfloor y_n \rfloor$.

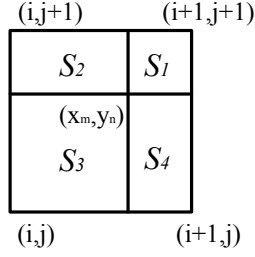


Figure 1: This figure shows the position of the corresponding non-distorted point and the integer points around them. These integer points are defined on the center of a pixel.

These areas can be calculated using the following codes,

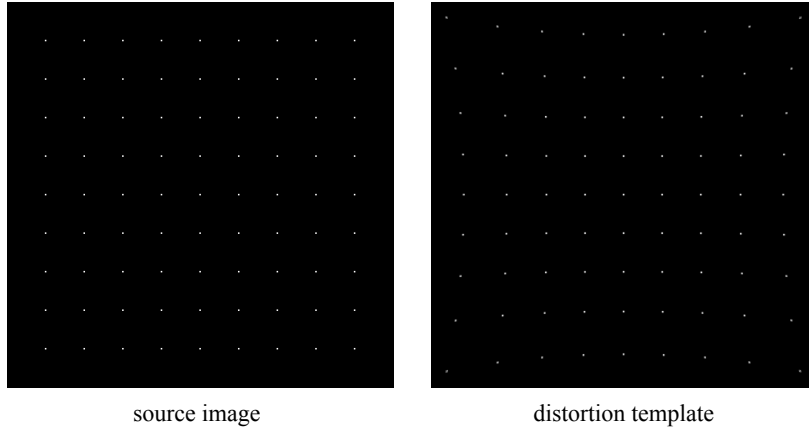
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1 s1 = (x_z+1-x_d).*(y_z+1-y_d);
2 s2 = (x_d-x_z).*(y_z+1-y_d);
3 s3 = (x_z+1-x_d).*(y_d-y_z);
4 s4 = (x_d-x_z).*(y_d-y_z);

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where $[x_z, y_z] = [i, j]$, and $I_d(m, n) = S_1 I(i, j) + S_2 I(i + 1, j) + S_3 I(i, j + 1) + S_4 I(i + 1, j + 1)$. The distorted image $I_d(m, n)$ can be then calculated, and the reconstruction method can be applied as well.

Based on the method of generating a distorted image, we can directly search for the parameter space for an optimal polynomial parameter. The input source image and the distortion template image is shown as follows.



The reconstruction method is to minimize the value function defined by

$V(k_1, k_2) = \sum_{m,n} (I(m, n) - I_r(m, n))^2$, where $I_r(m, n)$ is calculated by parameter k_1, k_2 .

A Matlab program is written to find the parameter. The reconstruction results are as follows.

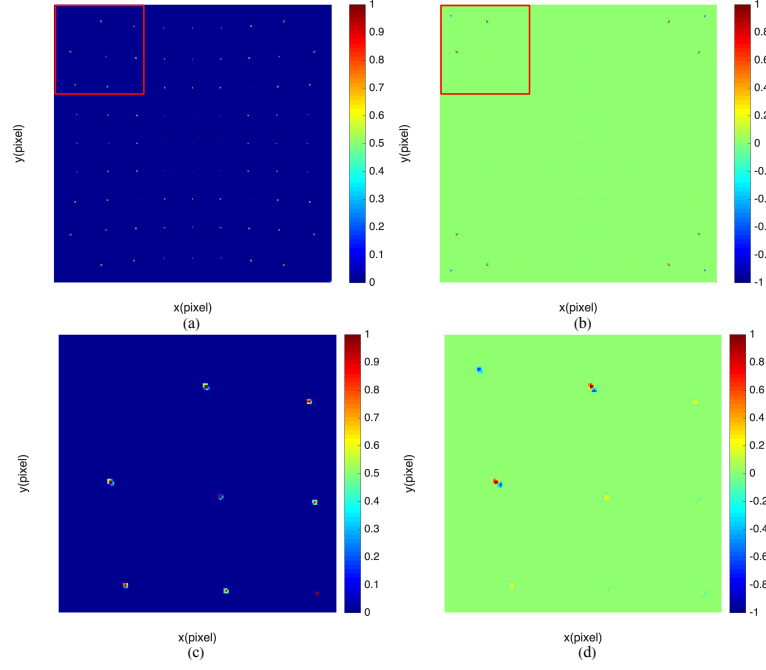


Figure 2: This figure shows the results of the pinhole model reconstruction. The reconstructed parameter is $k_1 = 1.81 \times 10^{-7}$, $k_2 = 1.50 \times 10^{-11}$. Figure (a) shows the reconstruction results. Figure (b) shows the error map of the reconstruction and the distortion template. Figure (c) and (d) are details at the corner of the field, marked in the red rectangle areas in figure (a) and (b).